5. STATISTICS

* Mean ($\bar{x}$)

→ Mean for ungrouped data:-

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\Sigma x_i}{n}$$

→ Mean for grouped data (Discrete freq. dist.)

$$\bar{x} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{f_1 + f_2 + \dots + f_n}$$

$$\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{\Sigma f_i x_i}{n}$$

where, $\Sigma f_i = n$ = total frequency.

→ continuous frequency Distri.

$$\bar{x} = A + \frac{\Sigma f_i d_i}{n} \times C$$

where, A = Assumed mean

C = class width

$n = \Sigma f_i$ = total freq.

$d_i = \frac{x_i - A}{C}$

* Find mean of 40, 42, 35, 28, 26, 39.
* Find mean of observations 39, 23, 58, 47, 50, 16 and 61.
* Find mean for the data 23, 20, 21, 17, 20, 23, 18, 19.
* Find mean of first eight Natural numbers.
* If mean of 4, 7, 6, a, 5, 9 is 6 then find value of 'a'.
* If mean of 10, 8, 5, x, 7 is 7 then find value of 'x'.
* Find mean of following data

x_i	0	1	2	3	4	5
f_i	3	20	15	8	3	1

* Find mean of following data

x_i	92	93	97	98	102	104
f_i	3	2	2	3	6	4

* Find mean of following freq. distribution.

Class	0-10	10-20	20-30	30-40	40-50
freq.	5	8	15	16	6

* Find mean of following data

Class	21-25	26-30	31-35	36-40	41-45	46-50
freq.	8	10	24	30	12	16

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* **Range** :- R = biggest observation − Smallest observation.

* **Mean deviation from mean** :- ($\delta\bar{x}$)

→ ungrouped data

$$\delta\bar{x} = \frac{1}{n}\sum|x_i - \bar{x}|$$

where, $\bar{x} = \frac{\sum x_i}{n}$

→ grouped data

$$\delta\bar{x} = \frac{1}{n}\sum f_i|x_i - \bar{x}|.$$

where $n = \sum f_i$

$$\bar{x} = \frac{\sum f_i x_i}{n}$$

→ Find range of data 18, 20, 28, 15, 17, 22, 25, 34, 29, 32.

* Find the mean deviation from mean for the data
37, 70, 48, 50, 32, 56, 63, 46, 54, 44.

* Find the mean deviation from mean for the data
58, 52, 45, 48, 57.

* Find the mean deviation from mean for the data

x_i	3	9	17	23	27
f_i	8	10	12	9	5

* Find mean deviation from mean of data

Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70
freq.	4	6	10	20	10	6	4

* Find mean deviation from mean of data

class	0-10	10-20	20-30	30-40	40-50
freq.	1	2	4	2	1

* Find mean deviation from mean for data

x_i	10	30	50	70	90
f_i	4	24	28	16	8

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* Standard deviation :-

* Ungrouped data

$$S = \sqrt{\frac{\Sigma (x_i - \bar{x})^2}{n}}$$

where $\bar{x} = \frac{\Sigma x_i}{n}$

* Grouped data

$$S = \sqrt{\frac{\Sigma f_i (x_i - \bar{x})^2}{n}}$$

where, $n = \Sigma f_i$

$\bar{x} = \frac{\Sigma f_i x_i}{n}$

* Find S.D. from the following data
6, 7, 10, 12, 13, 4, 8, 12.

* Find S.D. for the data
120, 132, 148, 136, 142, 140, 165, 153

* Find S.D. for the data

x_i	6	7	8	9	10	11	12
f_i	3	6	9	13	8	5	4

* Find S.D. for the data

x_i	4	8	11	17	20	24	32
f_i	3	5	9	5	4	3	1

* Find S.D. for following data

Class	0-10	10-20	20-30	30-40	40-50
freq.	5	8	15	16	6

* Find S.D. for the data

Class	0-9	10-19	20-29	30-39	40-49	50-59	60-69	70-79
freq.	6	14	16	24	20	10	6	4

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